RESUME

GUHANKARTHIK A G

B.EElectrical and Electronics Engineering

9994246331

Guhan.karthik25@gmail.com

LinkedIn: www.linkedin.com/in/guhankarthik-a-g-86379a2a1

GitHub: https://github.com/Ghnkrk

Objective:

Electricaland Electronics Engineering student with strong interest in machine learning, statistics, and modern AI systems. Experienced in building and training Transformer-based language models, working with LLM fine-tuning, RAG, and agentic systems. Focused on developing a solid end-to-end understanding of ML systems, from theory to deployment.

Education:

B.E Electrical and Electronics Engineering 2023-2027
(Minors in Data Science)
Kumaraguru College of Technology **-** CGPA 7.9
12-th Boards (Comp. Sci)
Vivekam mat. hr. sec. school **-** Percentage 93

Soft Skills:

Analytical thinking
Logical decision making
High adaptability
Calm under pressure
Efficiency-driven

Technical Skills:

Programming:
Python, C++
AI / ML:
Machine Learning, Deep Learning, Transformers & Large Language Models (LLMs), LLM Training & Fine-Tuning (LoRA, Adapters), Retrieval-Augmented Generation, Agentic Systems (ReAct, Multi-Agent Concepts), Quantization, On-Device Inference
Databases:
PostgreSQL, Vector Databases (FAISS, Qdrant)
Tools & Frameworks:
PyTorch, Hugging Face (Transformers, Datasets, PEFT), LangGraph, LangSmith (basic), llama.cpp, Weights & Biases, Streamlit, scikit-learn, TensorFlow (basic), GitHub
Version Control:
Git (basics), DVC (basics)
Others:
Docker (basics), Linux

Projects

MedComply – Deterministic AI Healthcare Compliance Reasoning Engine
(Huggingface, NLTK, groq)

- Built a rule-based engine to perform clause-level NABH compliance analysis from hospital SOPs using structured archetypes and token-normalized evidence matching.
- Integrated controlled semantic validation and LLM-assisted recommendations to generate traceable compliance gap reports.

TinyLLM–Training Auto-regressive Transformer Language Models - (pytorch, weights and biases)

- Built and trained light weight decoder-only Transformer models (2M–20M parameters) from scratch to study modern LLM architectures, training dynamics, and inference behavior.
- Implemented components including RoPE, SwiGLU, LoRA/adapters, and step-based training with monitoring and evaluation.

AutoAgenticML – Agent-Driven Machine Learning Automation Framework
(Langgraph, docker, groq, scipy, scikit-learn)

- Designed a multi-agent orchestration system to coordinate data ingestion, preprocessing, model selection, training, and evaluation with minimal manual intervention.
- Built decision-based control loops and modular task agents to enable structured, autonomous ML pipeline management.

On-Device LLM Inference Using LoRA and GGUF
(huggingface, peft, llama.cpp, termux)

- Fine-tuned a Qwen2.5-3B model using LoRA on instruction datasets and deployed it fully offline on an Android device using GGUF and llama.cpp.
- Analyzed inference latency and performance on mobile hardware.

Internship:

MachineLearning Engineer Intern - The Fusion Apps

Learning, Research on Vision models main focus in Facial detection, Facial recognition, Enhancement, Facial Quality Analysis and Filtering. Trained an experimental violence detection model with MultiViTV2 + BiGRU. Explored Object detection, recognition. Proposed an intermediary Facial quality check layer using landmark models for minimizing duplicate identities in facial recognition pipeline. Researched how SOTA facial recognition methods like Partial FC, Subcentre-ArcFace performs.

Certification:

-AWSAcademy Graduate – AWS Academy Machine Learning foundations
-Programming for Everybody (Getting Started with Python) - Coursera
-Python Data Structures - Coursera

Personal Interest

Listening to classics
Football
Cycling
Reading webnovels

Languages Known

English, Tamil

DOB: 25-06-2005

Address : 7/2A1, no.4 Veerapandi,
Perumal nagar, Naickenoor, Press colony post, Coimbatore - 641019